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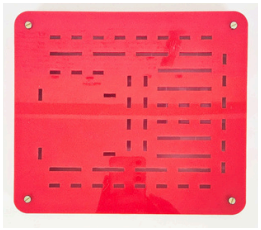
SESSION NAME

Session

2

STRUCTURE AND FASTENERS OF MICROCONTROLLERS

1


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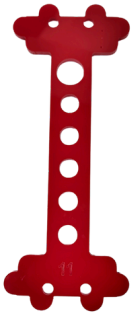


1

Spacers

2

L Clamp

3

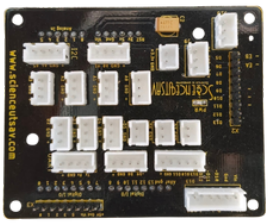
Spacer Screws

4

Nuts

5

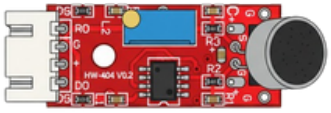
Screw

6

Arduino With Shield

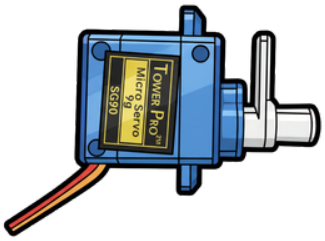
7

Arduino cable

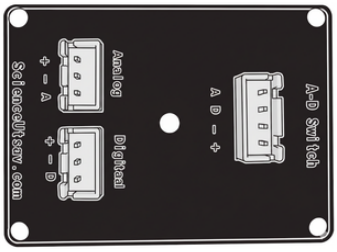
ELECTRONIC COMPONENTS



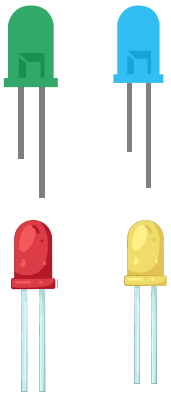
Sound Sensor – A 4-pin Sound Sensor is a module that uses a microphone to detect sound levels in the environment. It converts sound waves into electrical signals and provides both analog (AO) and digital (DO) outputs. The module has four pins: VCC, GND, AO, and DO. It is commonly used in projects such as clap switches, sound-activated lights, and noise detection systems.



Servo Motor – A Servo Motor is a small, high-precision motor used to control the position of an object accurately. It can rotate to a specific angle based on control signals from a microcontroller. A servo motor typically has three pins: VCC, GND, and Signal. It is widely used in robotics, automated systems, and projects that require precise movement, such as robotic arms and steering mechanisms.



AD switch PCB- An AD switch PCB is a small circuit board that contains a switch along with basic electronic components to provide both analog and digital signal outputs. It is typically used to detect button presses or toggle actions and send signals to a microcontroller. The PCB may include pins for VCC, GND, analog output (AO), and digital output (DO), making it useful in projects where both precise (analog) and simple on/off (digital) input detection is needed.



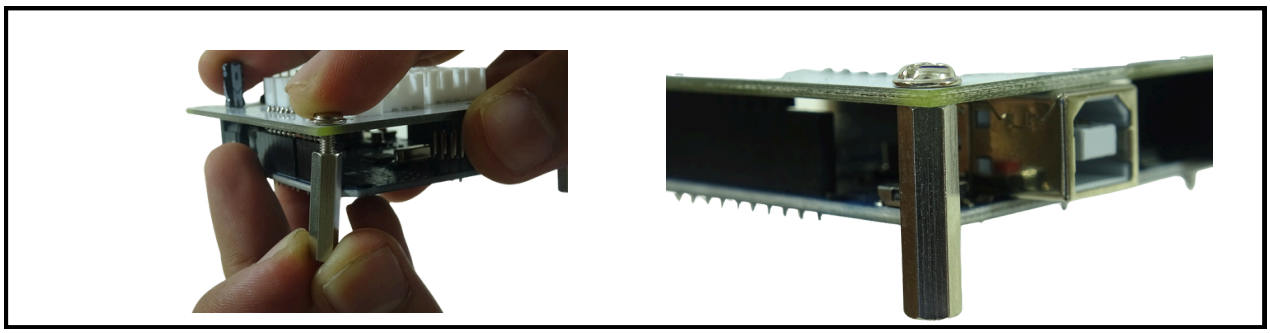
DIP LED- A DIP LED (Dual In-line Package Light Emitting Diode) is a two-pin electronic component that emits light when electric current flows through it. It is widely used as an indicator and light source in electronic circuits because of its low power consumption, long life, and high efficiency.

2 PIN RMC

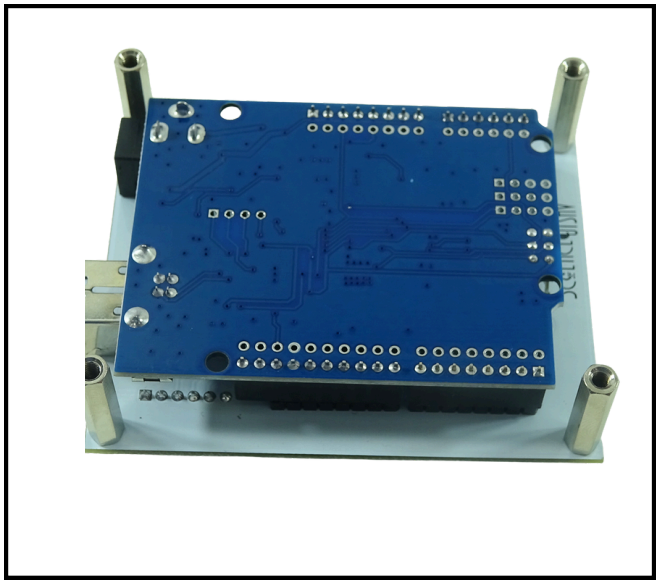
Used to connect 1-watt LEDs, Buzzer, DIP LEDs

Procedure :

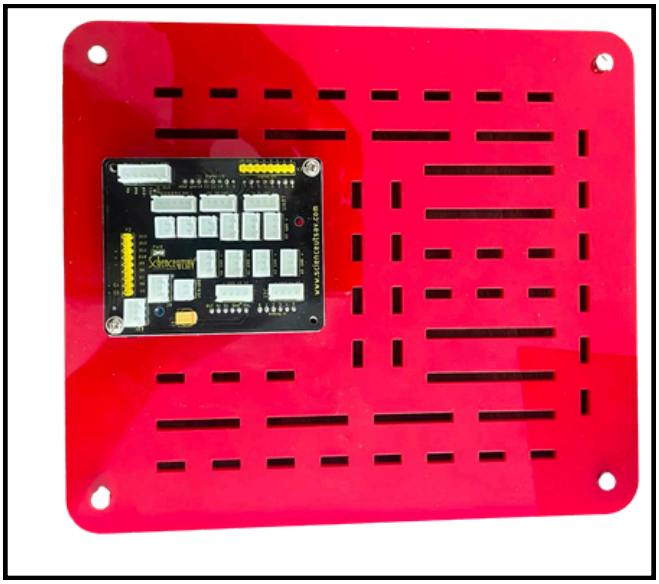
Step 1 : Fit the spacer to Arduino Shield as shown below using Spacer screw



Step 2 : Repeat the step 1 for all 4 corners of Arduino Shield



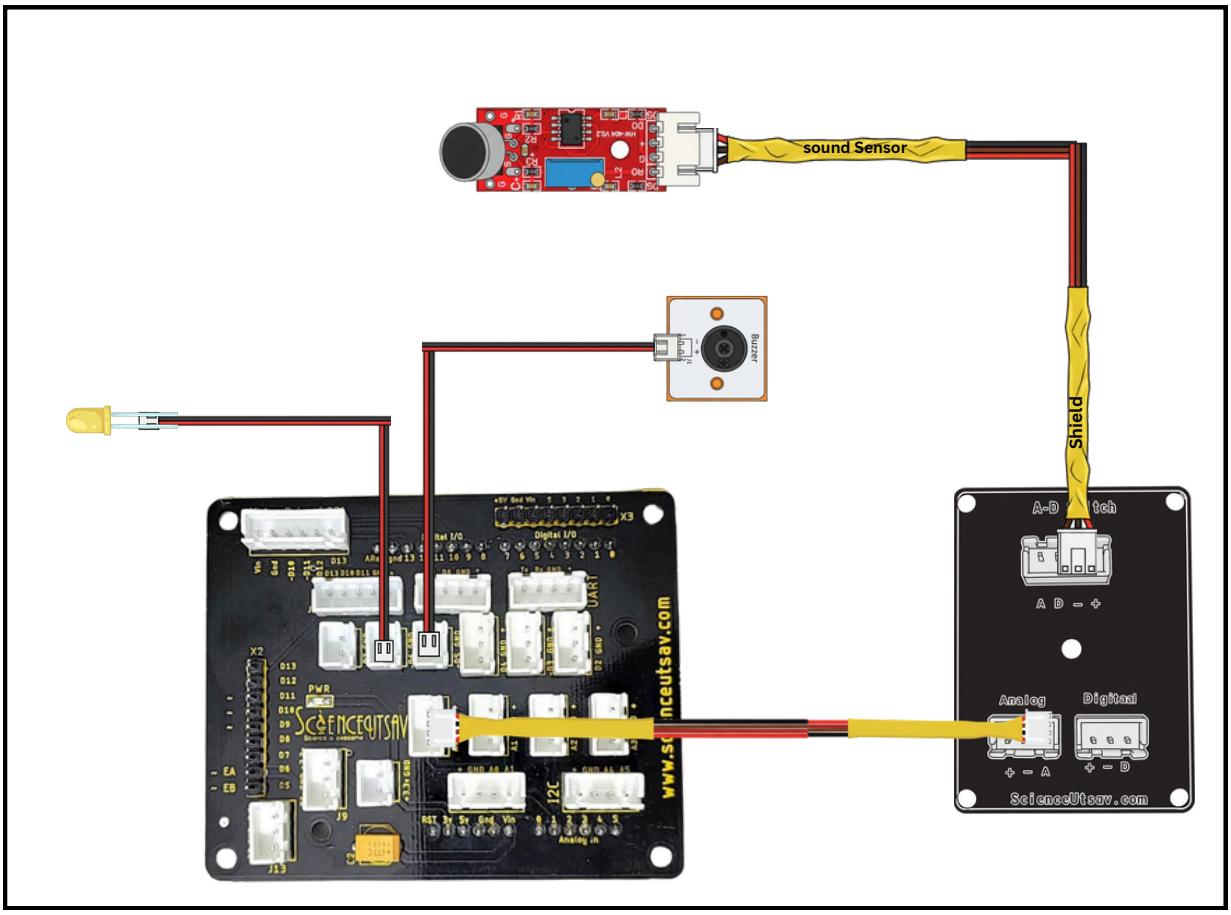
Step 3 : Place the Arduino Shield to the base board as shown below



Step 4 : Fix it with the spacer screw from the bottom as shown below

BEGINNER PROJECT

The circuit diagram of the project is given below

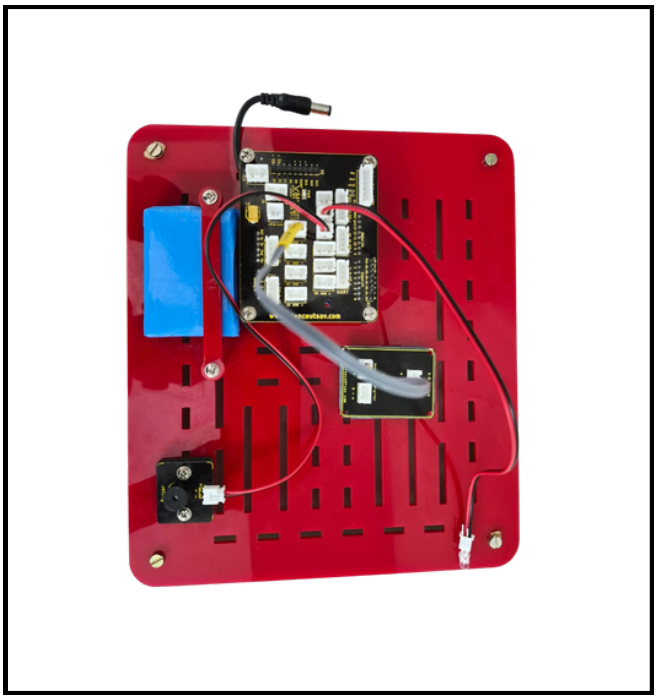


CONNECTION:

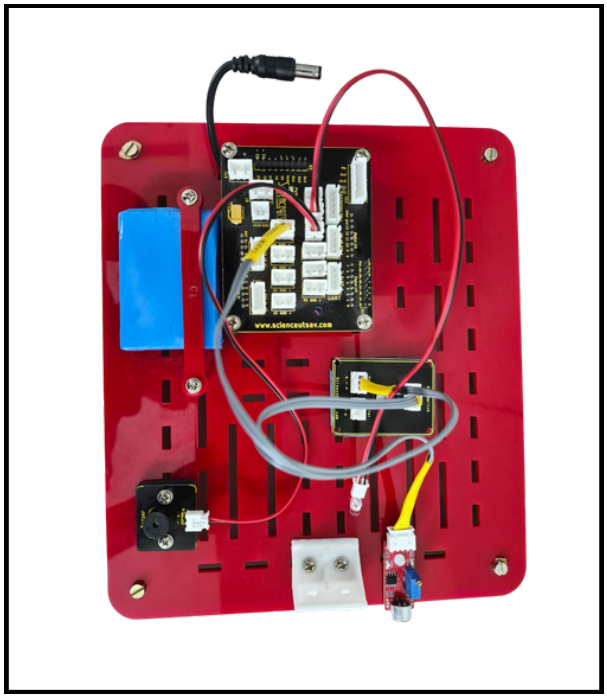
1. Dip LED- D6
2. Buzzer- D5
3. AD switch- A0

Step 1- Connect the buzzer and dip LED as shown

Step 2- Attach the AD switch to the board

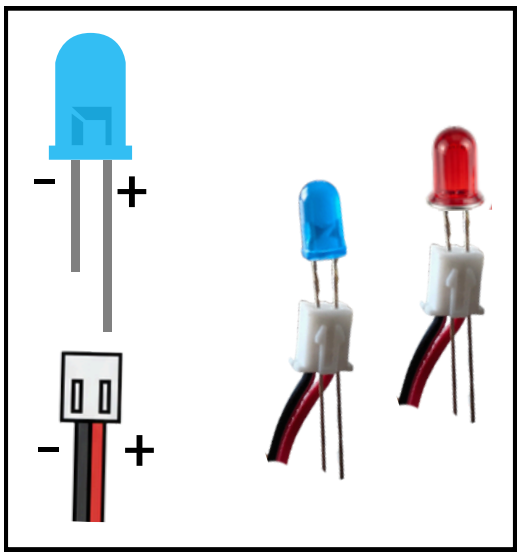


Step 3- Connect the sound sensor to the AD switch



Step 4- Attach the LED to the template as shown

Insert the LEDs from one side , long lead (+)of led to the red wire and short lead(-) of led to black wire



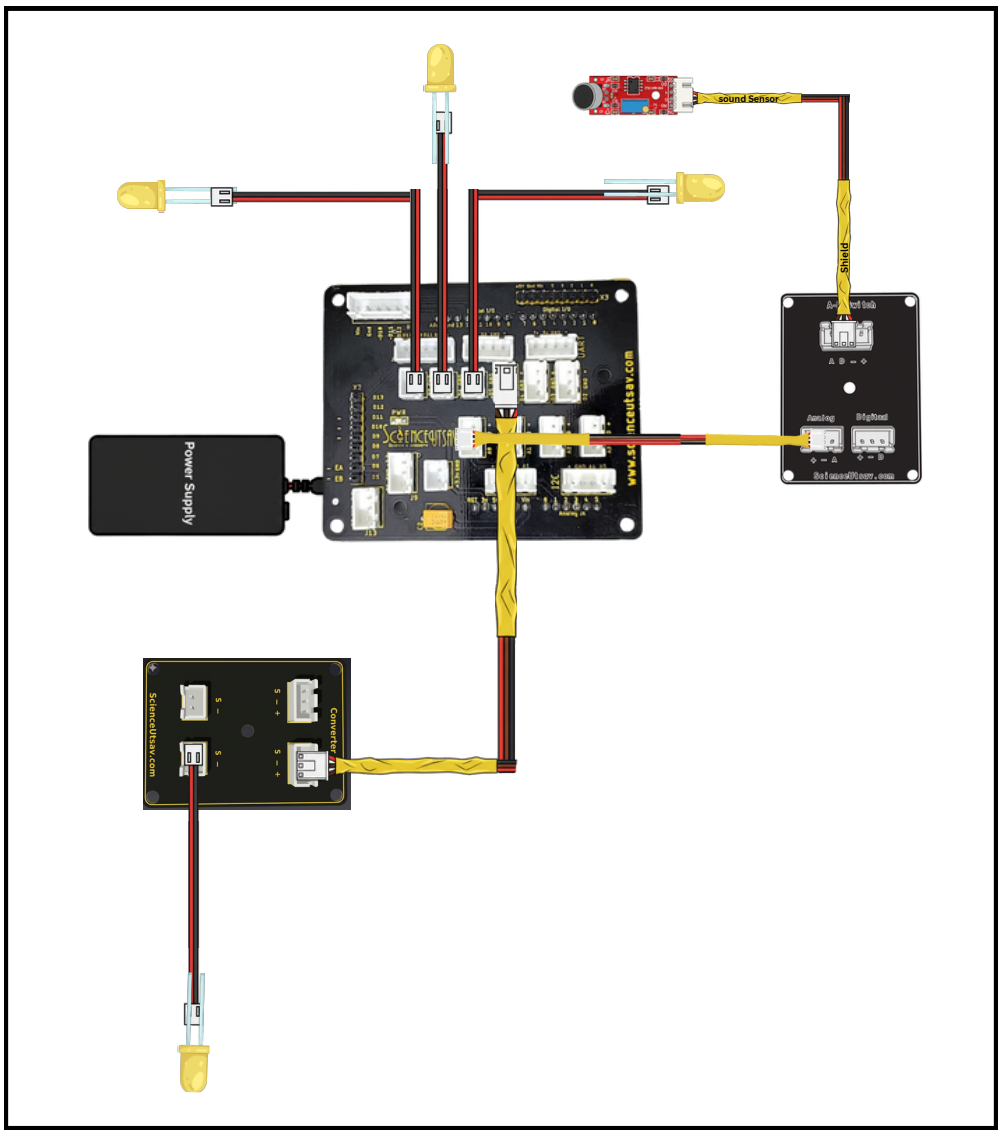
Step 5- Attach the template to the L-clamp and fix it to the board

Code

```
// Session 2 Grade 6
// Birthday Party / Cake
// Sound Sensor + LED Only
// Blow / sound detected = LED OFF, then reset ON

const int soundSensor = A0;
const int led = 6;

int soundLimit = 600;  // adjust after testing

void setup() {
  pinMode(led, OUTPUT);
  digitalWrite(led, HIGH);  // candle LED ON initially
}

void loop() {
  int soundValue = analogRead(soundSensor);

  if (soundValue > soundLimit) {
    digitalWrite(led, LOW);   // candle OFF
    delay(3000);              // stay OFF for 3 seconds
    digitalWrite(led, HIGH);  // reset candle ON
  }
}
```

BEGINNER PROJECT

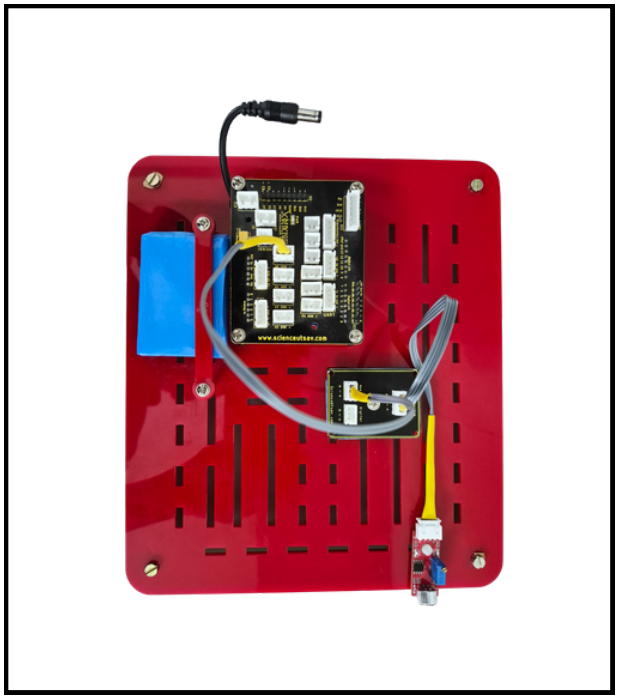
The circuit diagram of the project is given below



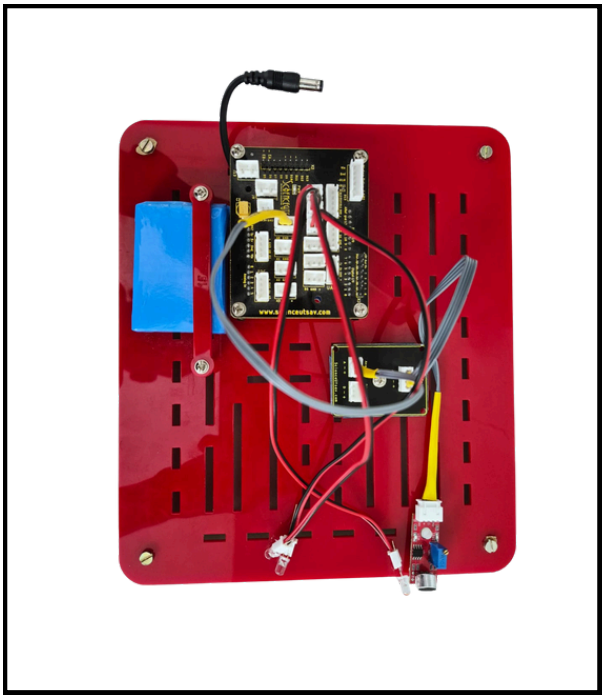
CONNECTION:

1. Dip LED- D5, D6, D7
2. Converter- D4
3. AD Switch- A0

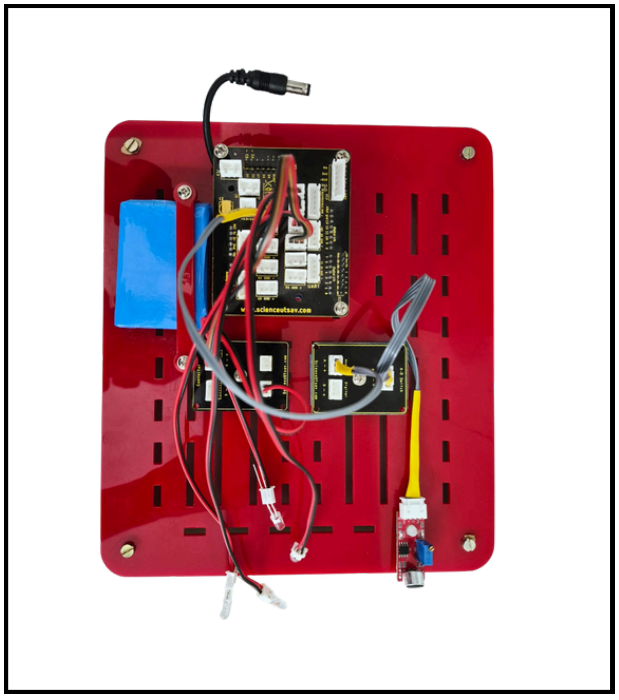
Step1- Connect the AD switch to the board and attach the sound sensor to it



Step 2- Fix the three dip LEDs to the board

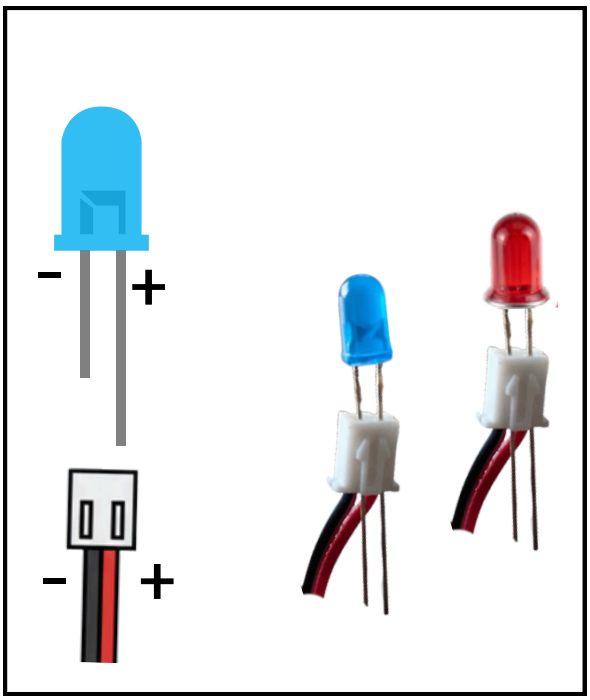


Step 3 - Connect the converter along with the fourth dip LED



Step 4- Attach the template to the board as shown

Insert the leds from one side , long lead (+)of led to the red wire and short lead(-) of led to black wire



Step 5- Fix the LEDs to the template as shown

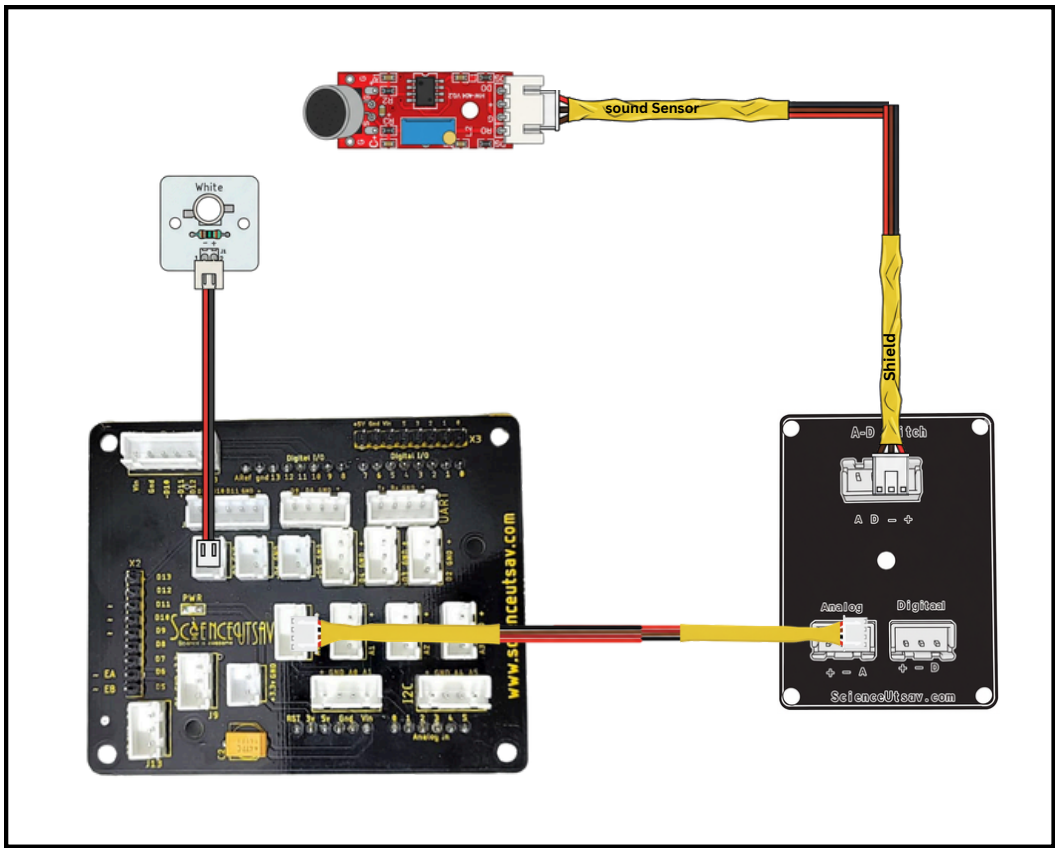
Step 6- Upload the code and observe

Code

```
// Session 2 Grade 6
// Birthday Party / Cake
// Sound Sensor + LED Only
// Blow / sound detected = LED OFF, then reset ON

const int soundSensor = A0;
const int led = 6;

int soundLimit = 600;  // adjust after testing

void setup() {
  pinMode(led, OUTPUT);
  digitalWrite(led, HIGH);  // candle LED ON initially
}

void loop() {
  int soundValue = analogRead(soundSensor);

  if (soundValue > soundLimit) {
    digitalWrite(led, LOW);   // candle OFF
    delay(3000);              // stay OFF for 3 seconds
    digitalWrite(led, HIGH);  // reset candle ON
  }
}
```

INTERMIDIATE PROJECT

The circuit diagram of the project is given below

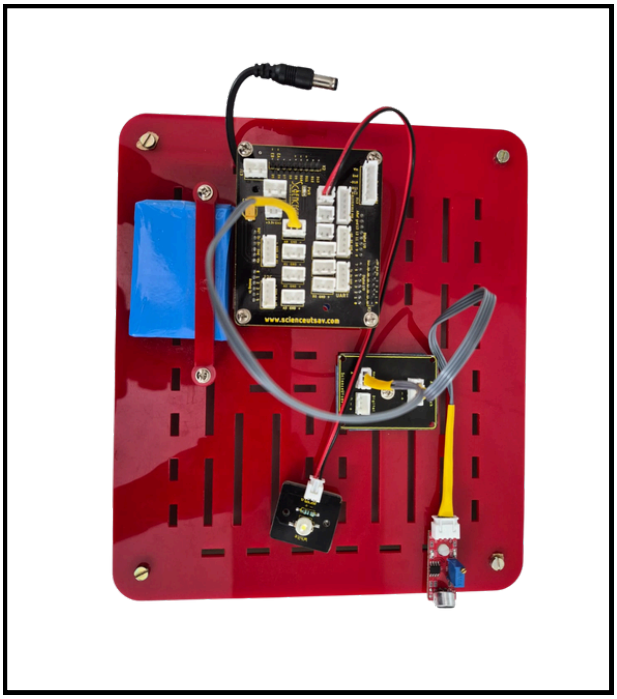


CONNECTION:

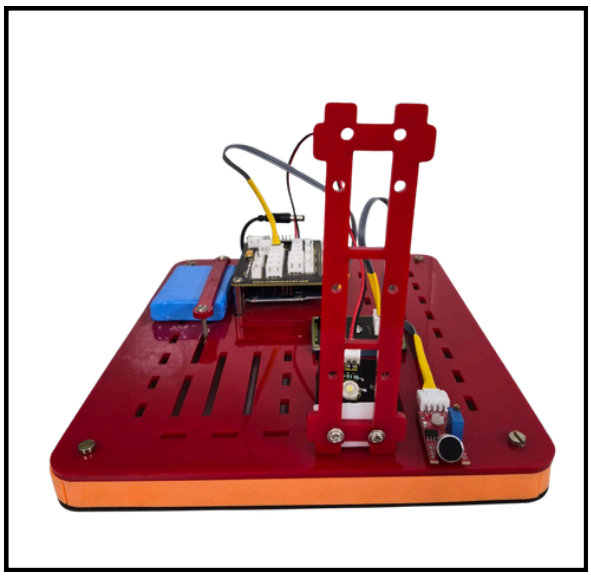
1. 1W LED- D7
2. AD switch- A0

Step 1- Connect the AD switch to the board and attach the sound sensor to it

Step 2- Connect the buzzer as shown



Step 3- Attach the stand to the board



Step 4- Attach template and fix the LED to it as shown

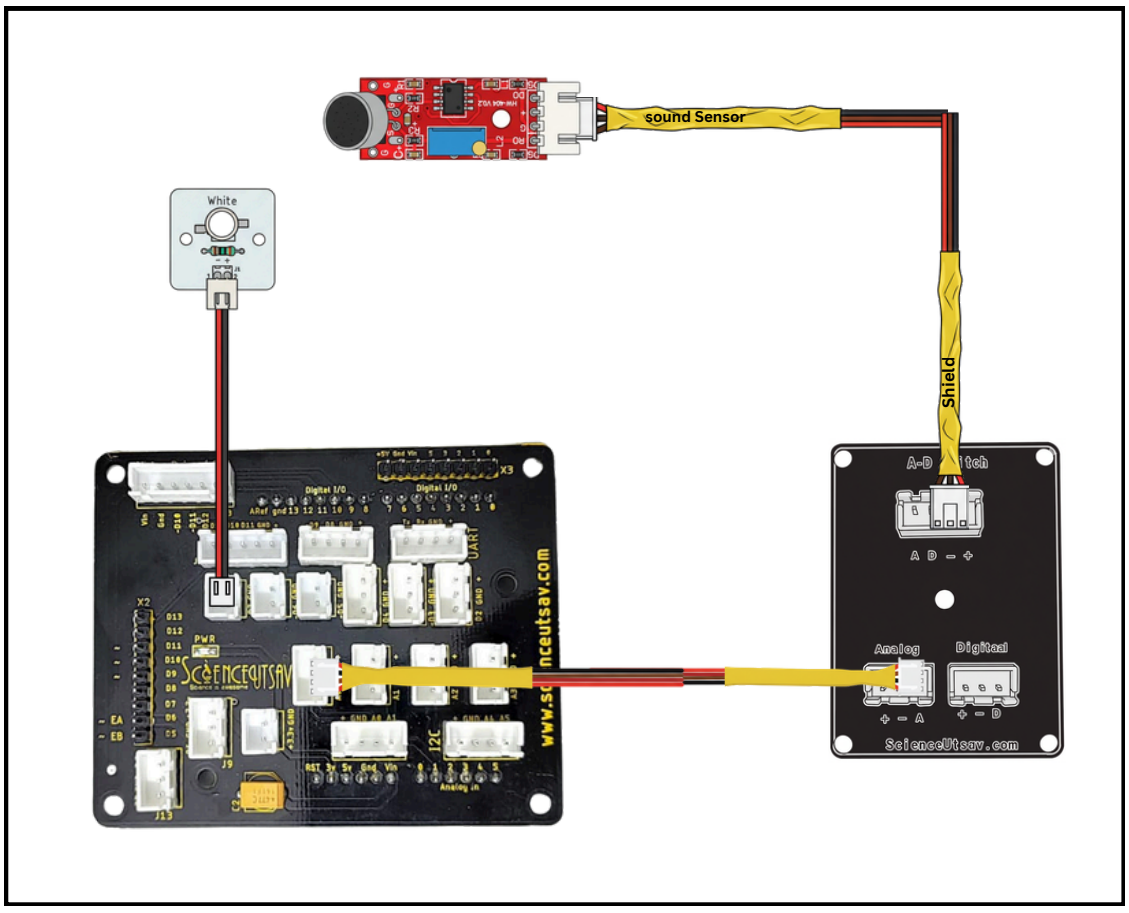
Step 5 - Attach the other template as shown

Code

```
// Session 2 Grade 7
// Clap Lamp / Lantern
// Clap once = LED ON, clap again = LED OFF

const int soundSensor = A0;
const int lampLED = 7;

int soundLimit = 600; // adjust after testing
bool lampState = false;
bool alreadyClapped = false;

void setup() {
 pinMode(lampLED, OUTPUT);
 digitalWrite(lampLED, LOW);
}

void loop() {
 int soundValue = analogRead(soundSensor);

 if (soundValue > soundLimit && alreadyClapped == false) {
 lampState = !lampState;
 digitalWrite(lampLED, lampState);
 alreadyClapped = true;
 delay(300);
 }

 if (soundValue < soundLimit - 100) {
 alreadyClapped = false;
 }
}
```

INTERMIDIATE PROJECT

The circuit diagram of the project is given below

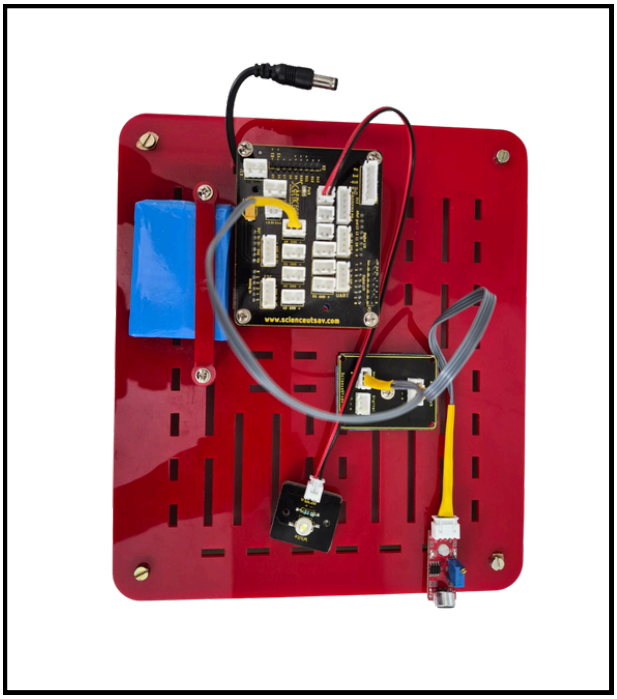


CONNECTION:

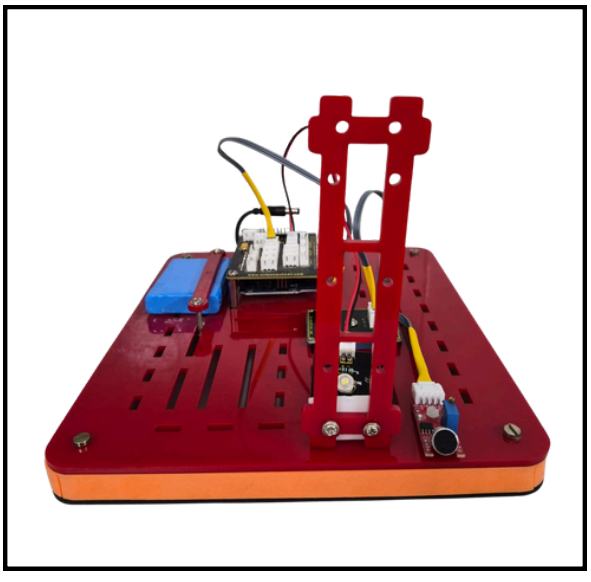
1. 1W LED- D7
2. AD switch- A0

Step 1- Connect the AD switch to the board and attach the sound sensor to it

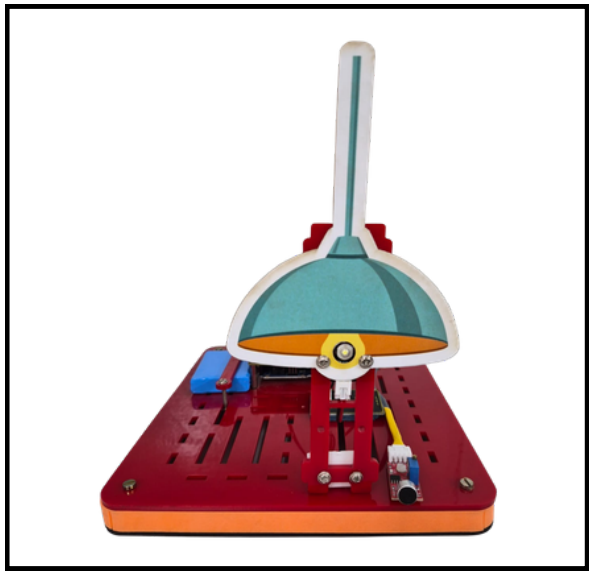
Step 2- Connect the buzzer as shown



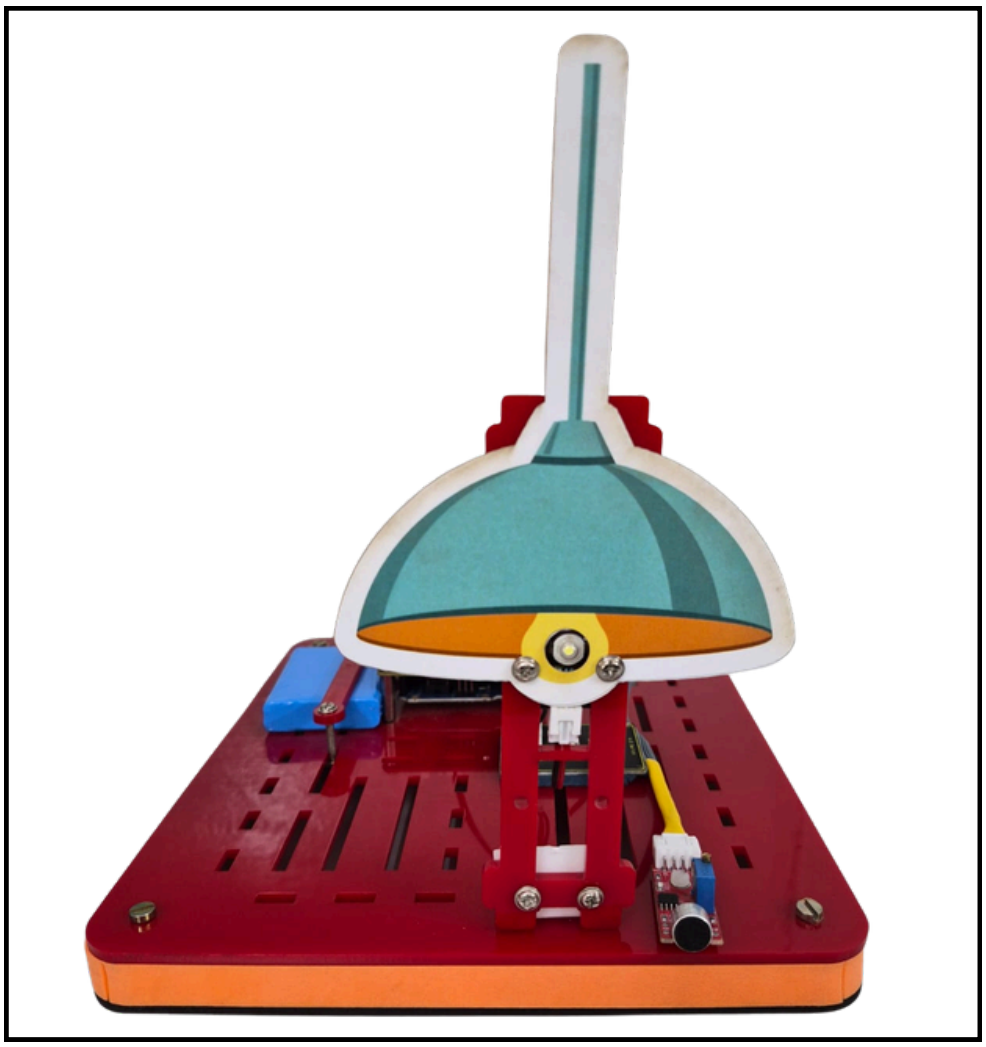
Step 3- Attach the stand to the board



Step 4- Attach template and fix the LED to it as shown



Step 6- Upload the code and observe

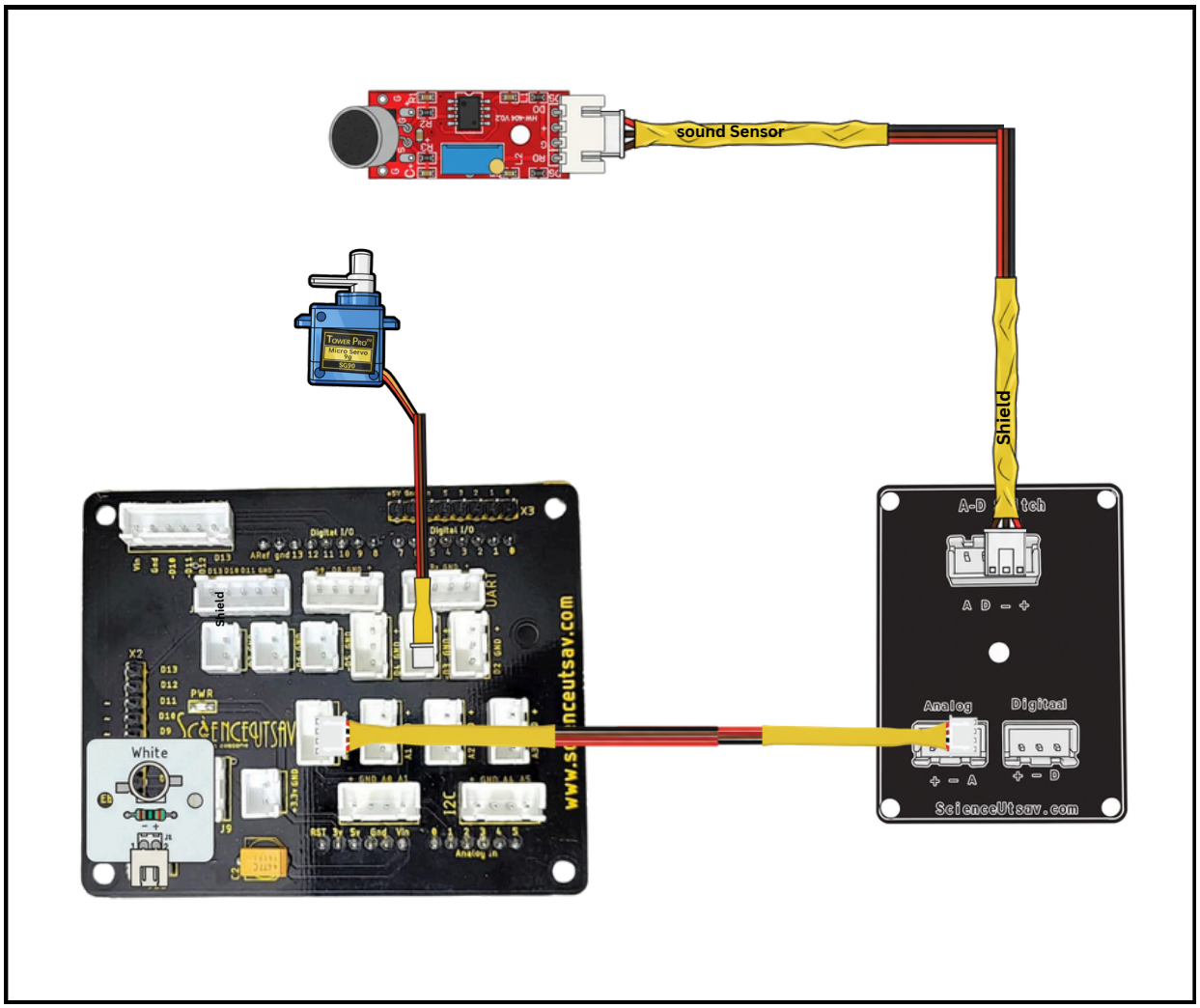


Code

```
// Session 2 Grade 7
// Clap Lamp / Lantern
// Clap once = LED ON, clap again = LED OFF

const int soundSensor = A0;
const int lampLED = 7;

int soundLimit = 600; // adjust after testing
bool lampState = false;
bool alreadyClapped = false;

void setup() {
 pinMode(lampLED, OUTPUT);
 digitalWrite(lampLED, LOW);
}

void loop() {
 int soundValue = analogRead(soundSensor);

 if (soundValue > soundLimit && alreadyClapped == false) {
 lampState = !lampState;
 digitalWrite(lampLED, lampState);
 alreadyClapped = true;
 delay(300);
 }

 if (soundValue < soundLimit - 100) {
 alreadyClapped = false;
 }
}
```

ADVANCE PROJECT

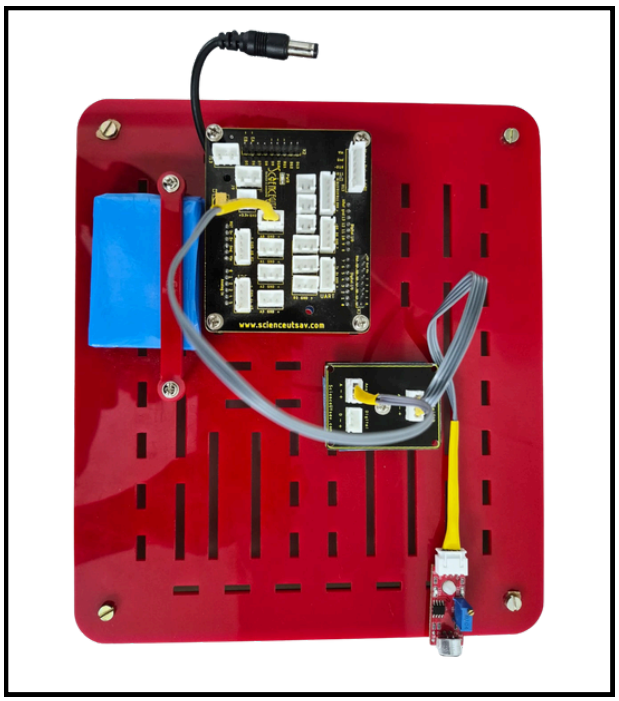
The circuit diagram of the project is given below



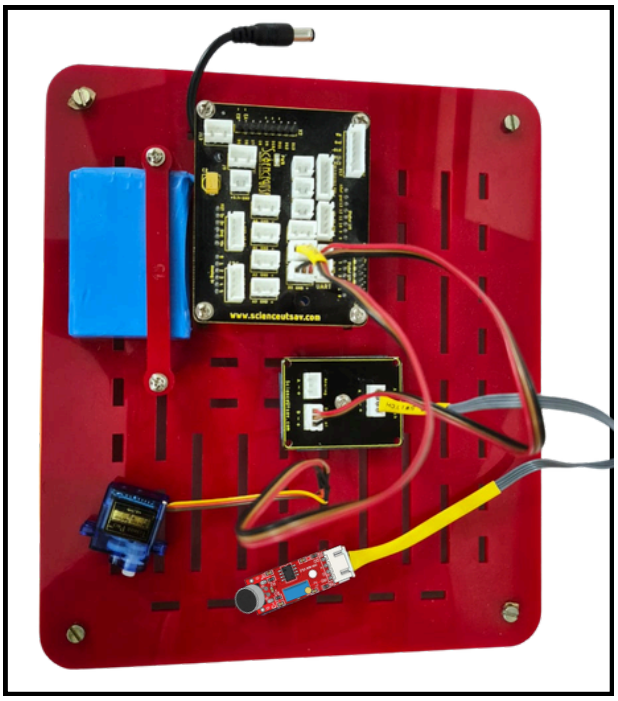
CONNECTION:

1. Servomotor- D3
2. AD switch- A0
3. Sound Sensor - AD switch - A0

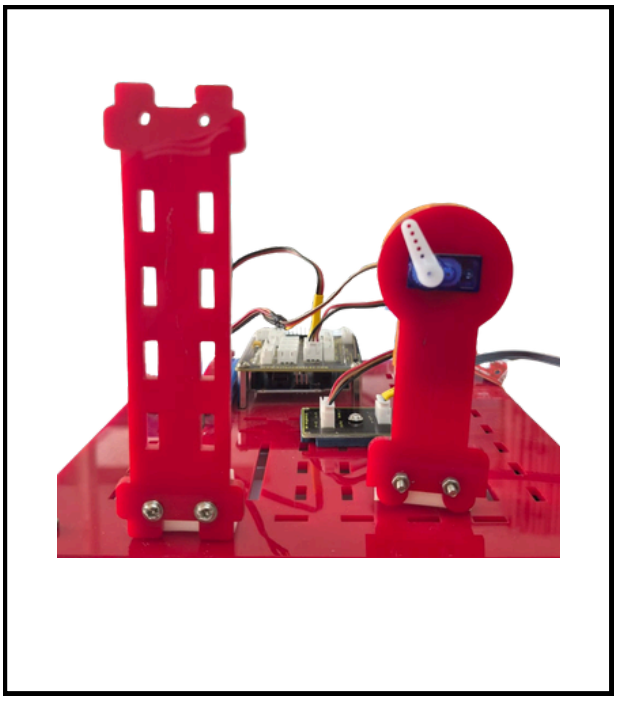
Step 1- Connect the AD switch to the board and attach the sound sensor to it



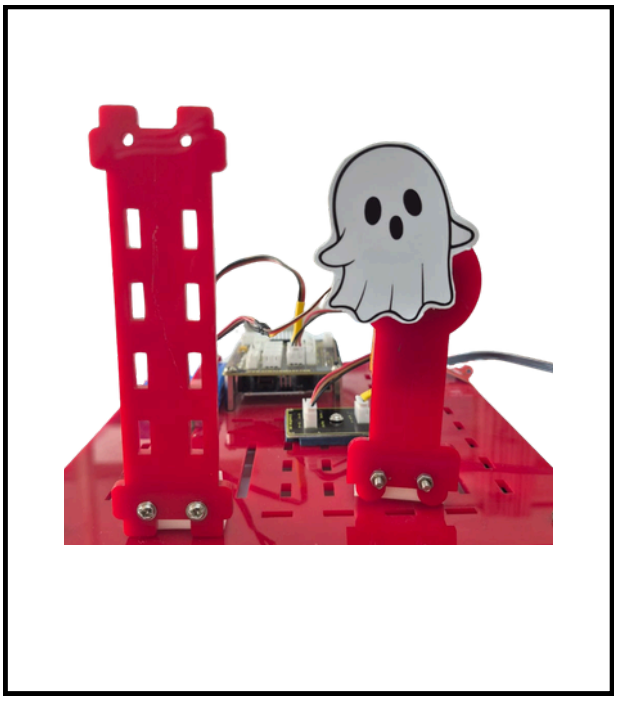
Step 2- Connect the the servomotor and halleffect sensor to the shield



Step 3- Attach the stands to the board as shown



Step 4- Attach the ghost template to the servomotor



Step 5- Attach the arch template as shown

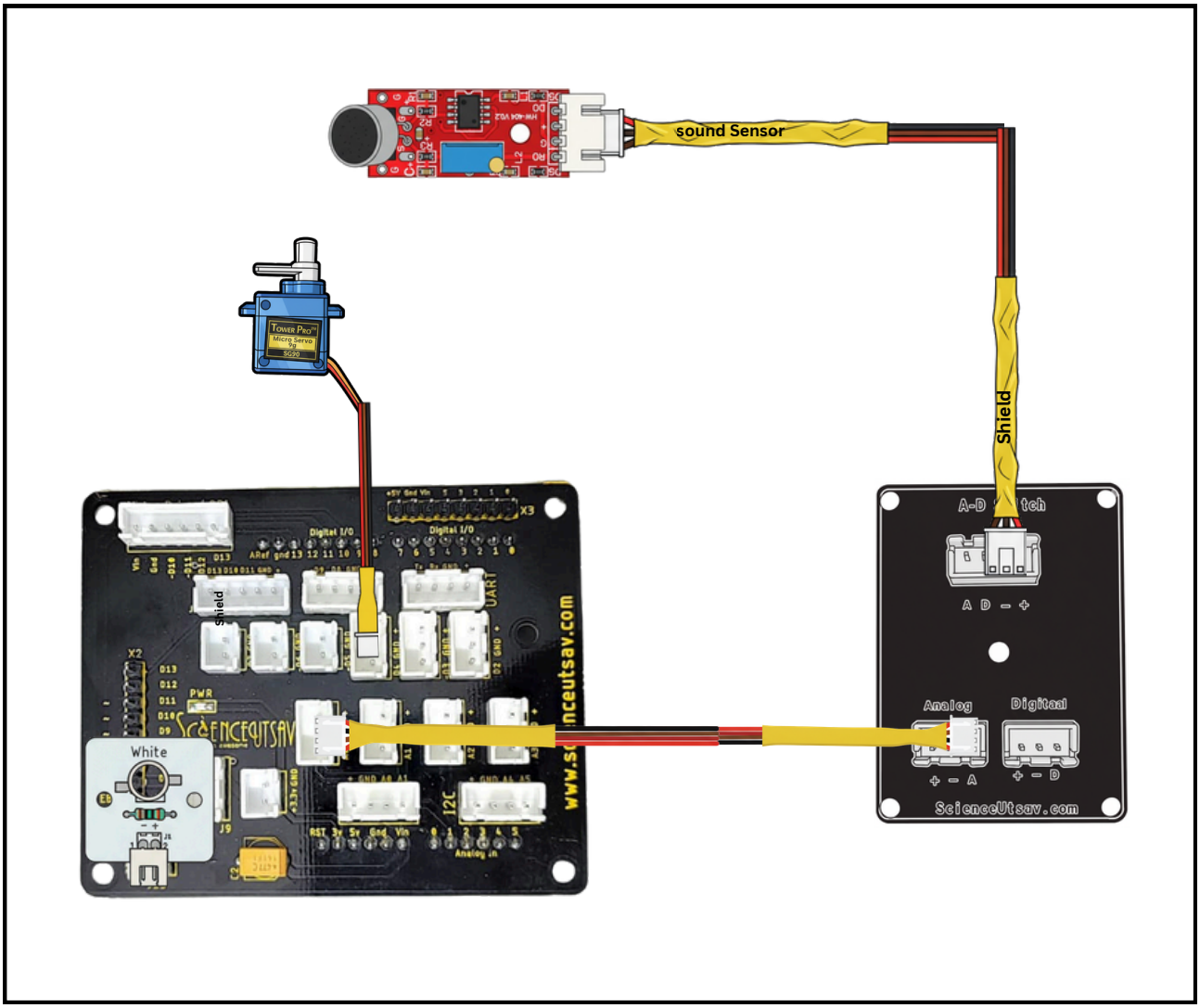
Step 6- Fix the sound sensor as shown

Code

```
#include <Servo.h>
Servo doorServo;

int irSensor = 2;
int servoPin = 3;

int IR_ACTIVE = LOW;   // Change to HIGH if opposite

void setup() {
  pinMode(irSensor, INPUT);
  doorServo.attach(servoPin);
  doorServo.write(130);   // Door closed
}

void loop() {

  if (digitalRead(irSensor) == IR_ACTIVE) {

    delay(1000);   // IR detected for 5 seconds

    // Open door smoothly
    for (int i = 130; i >= 30; i--) {
      doorServo.write(i);
      delay(40);

    }

    delay(5000);   // Door open for 5 seconds
    for (int i = 30; i <= 130; i++) {
      doorServo.write(i);
      delay(25);
    // Close door smoothly

    }
  }
}
```

ADVANCE PROJECT

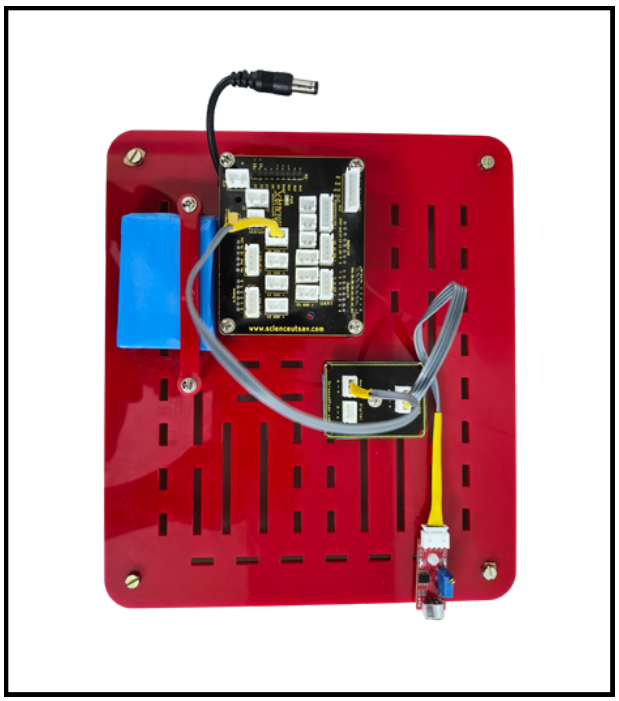
The circuit diagram of the project is given below



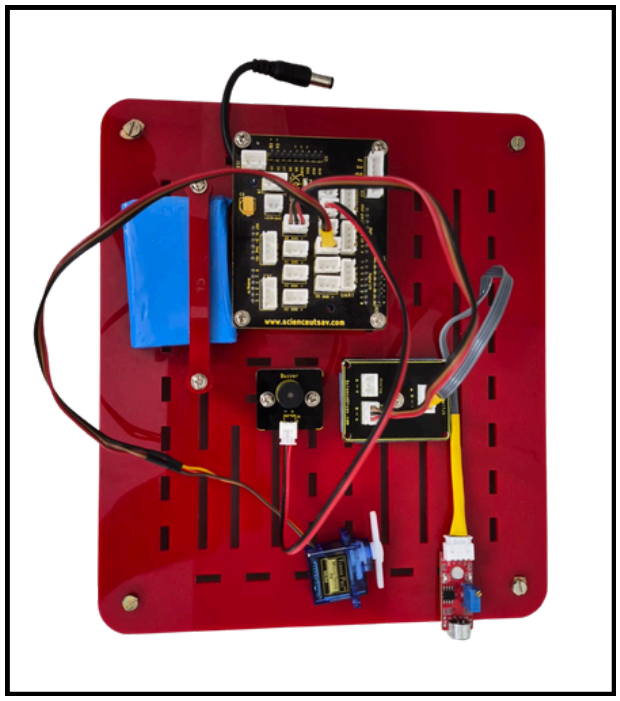
CONNECTION:

1. Servomotor- D4
2. AD switch- A0

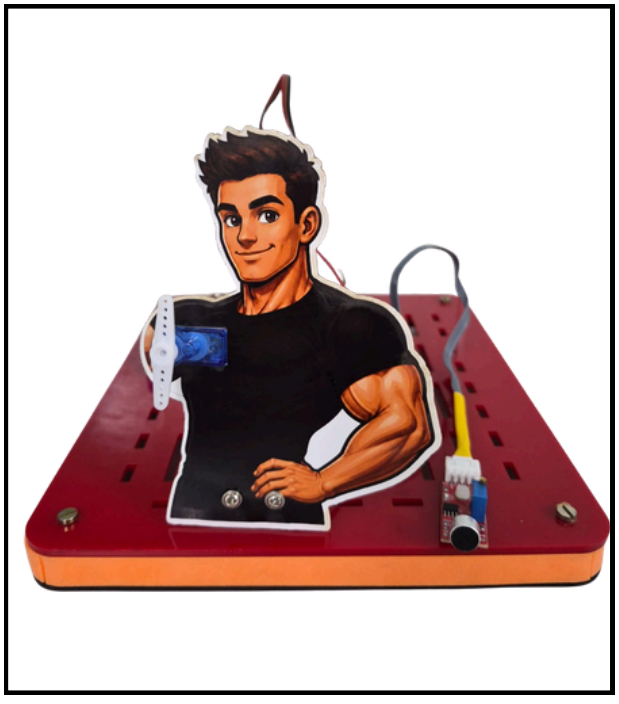
Step 1- Connect the AD switch to the board and attach the sound sensor to it



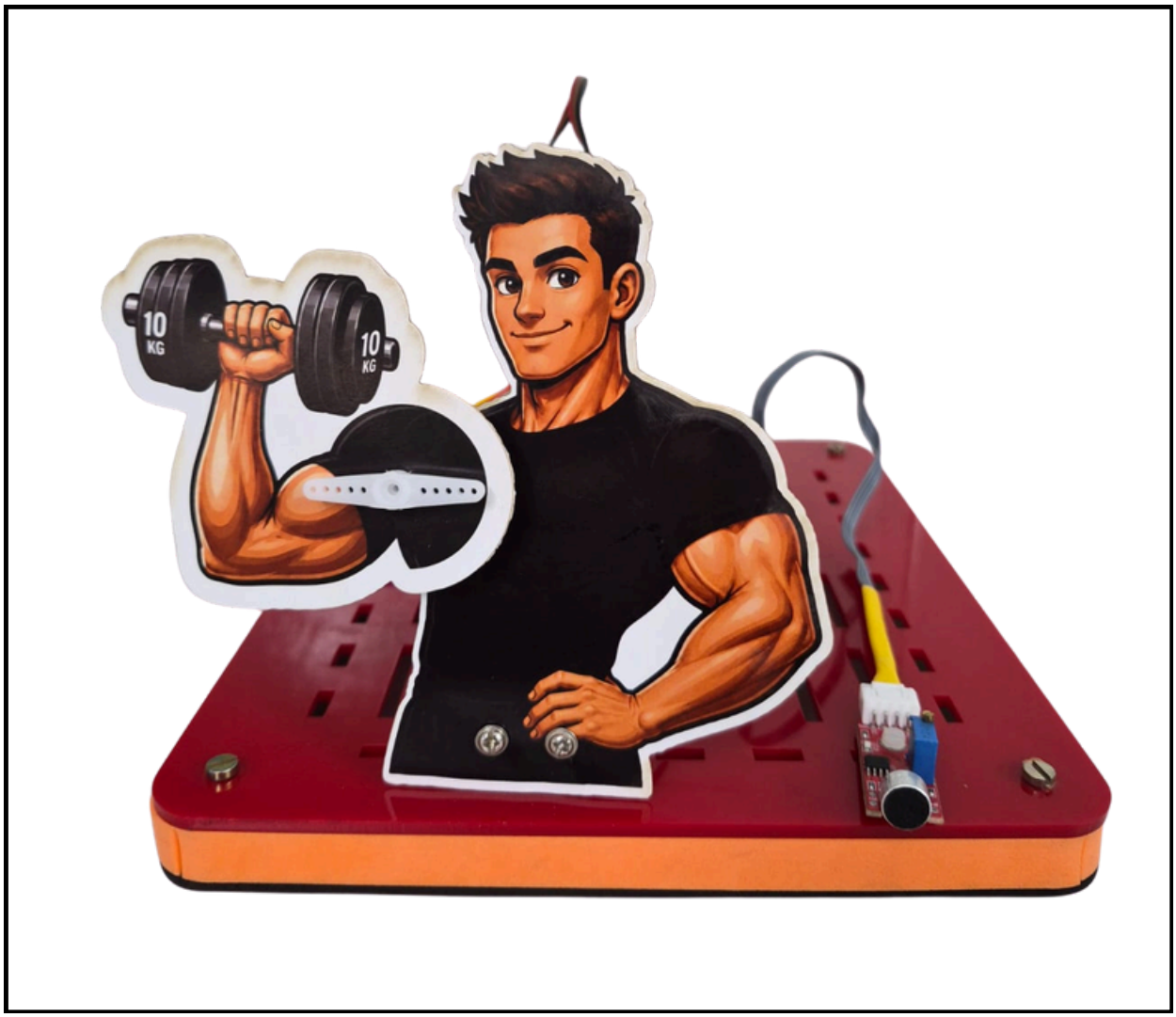
Step 2- Connect the servomotor and buzzer as shown



Step 3- Attach the template servomotor holder



Step 4- Attach the other template as shown


10
KG
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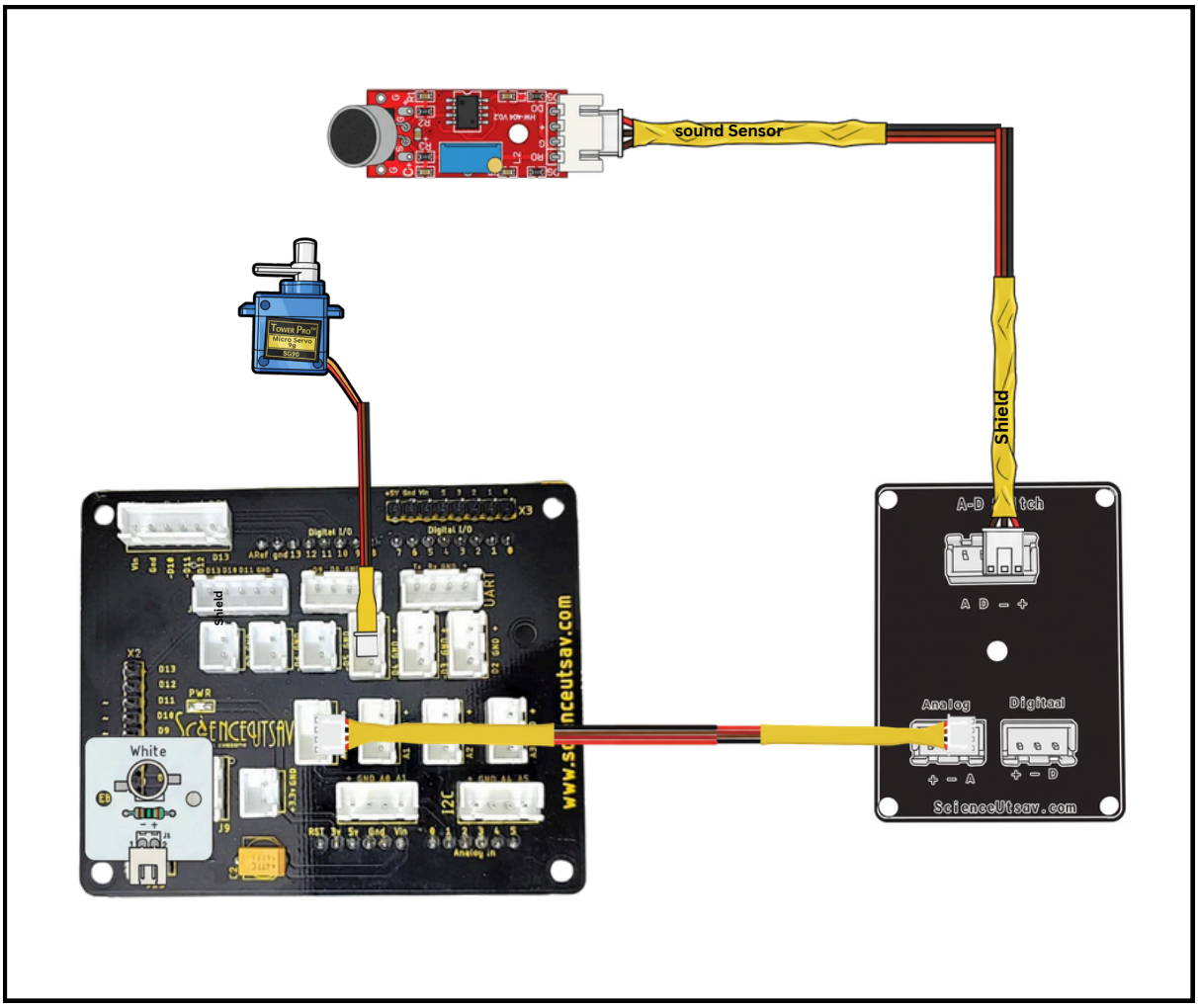
Code

```
// Session 2 Grade 8
// Body Builder
// Sound detected = servo moves arm like bicep movement

#include <Servo.h>

const int soundSensor = A0;
const int servoPin = 4;

int soundLimit = 600; // adjust after testing

Servo bodyServo;

void setup() {
 bodyServo.attach(servoPin);
 bodyServo.write(0);
}

void loop() {
 int soundValue = analogRead(soundSensor);

 if (soundValue > soundLimit) {
 for (int i = 0; i < 3; i++) {
 bodyServo.write(90);
 delay(400);
 bodyServo.write(0);
 delay(400);
 }
 }
}
```

EXPERT PROJECT

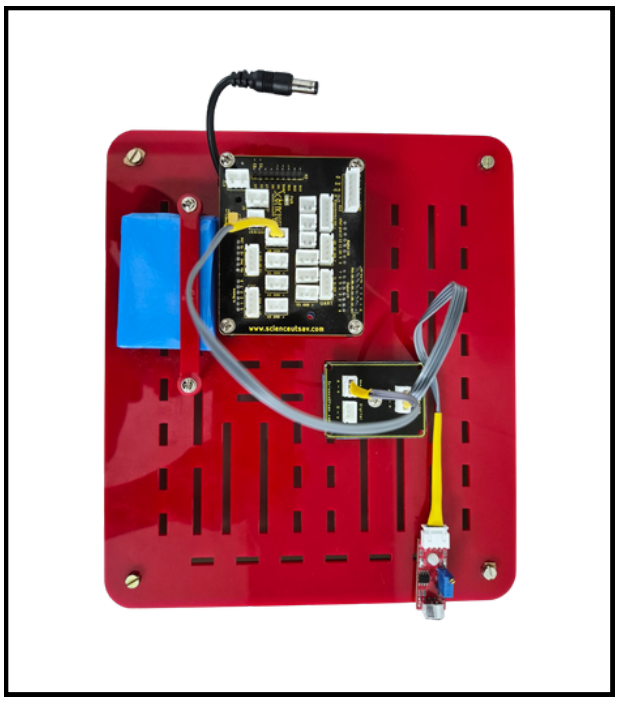
The circuit diagram of the project is given below



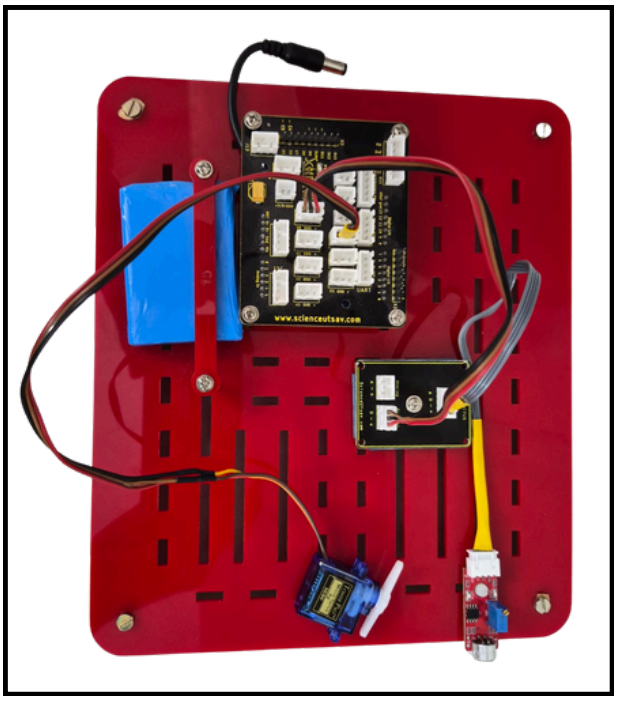
CONNECTION:

1. Servomotor- D4
2. AD switch- A0

Step 1- Connect the AD switch to the board and attach the sound sensor to it



Step 2- Connect the servomotor as shown



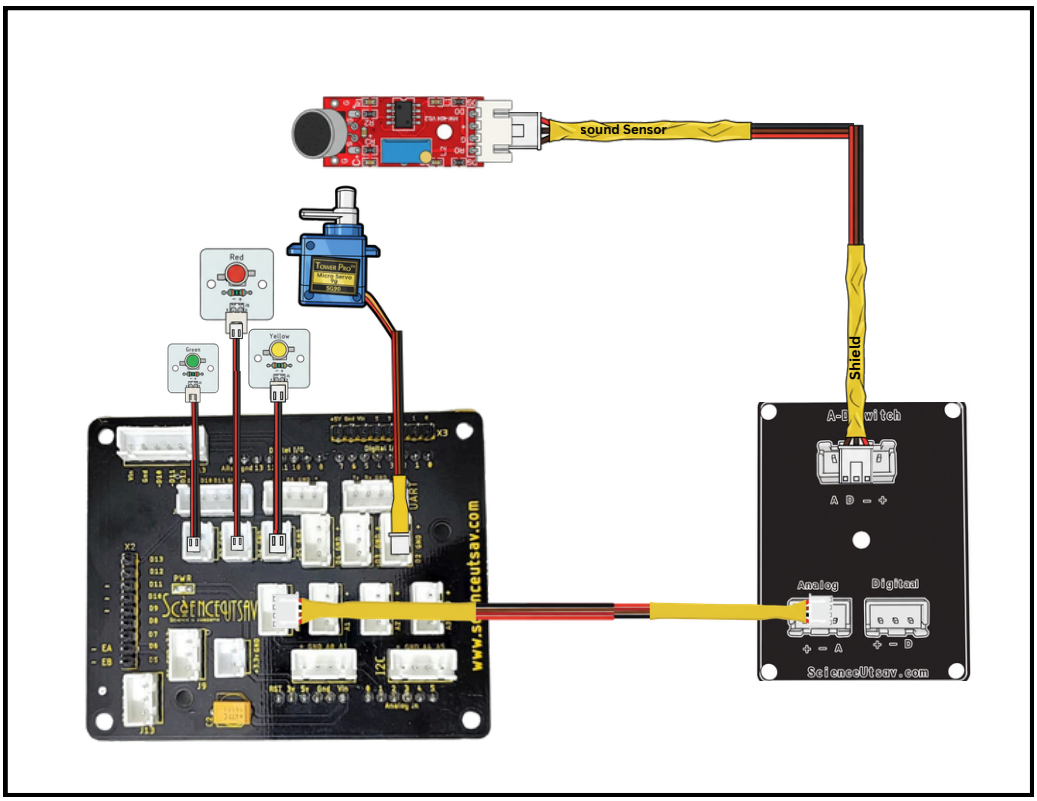
Step 3- Attach the template servomotor holder

Code

```
// Session 2 Grade 9
// Shout Meter
// More sound = servo moves more

#include <Servo.h>

const int soundSensor = A0;
const int servoPin = 4;

Servo meterServo;

void setup() {
 meterServo.attach(servoPin);
 meterServo.write(0);
}

void loop() {
 int soundValue = analogRead(soundSensor);

 int angle = map(soundValue, 300, 900, 0, 180);
 angle = constrain(angle, 0, 180);

 meterServo.write(angle);
 delay(100);
}
```

EXPERT PROJECT

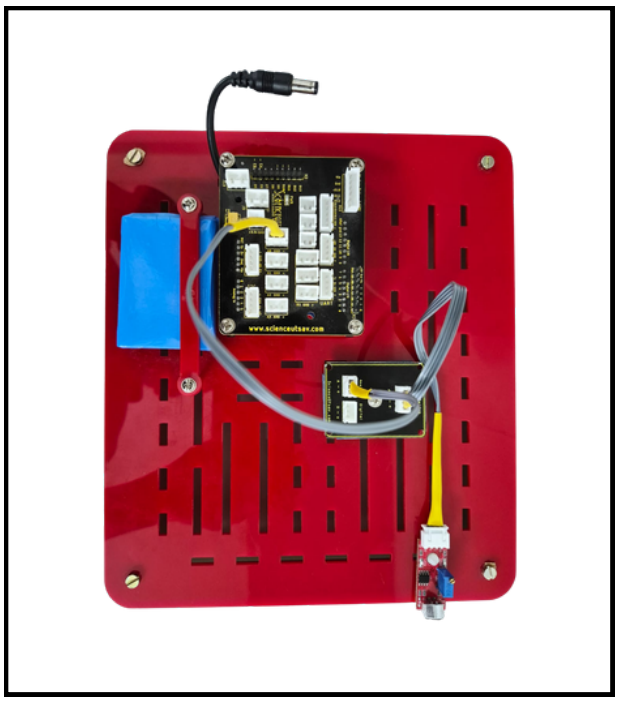
The circuit diagram of the project is given below



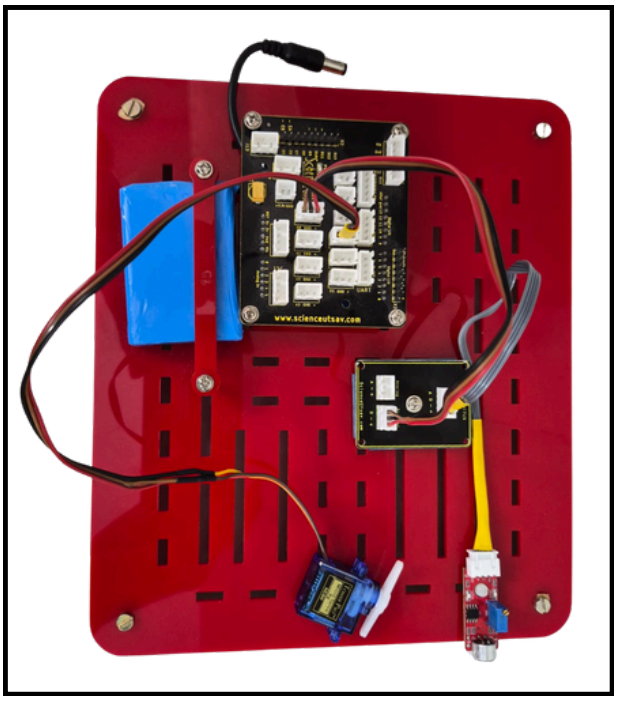
CONNECTION:

1. Servomotor- D2
2. AD switch- A0
3. Yellow 1W LED- D5
4. Red 1W LED- D6
5. Green 1W LED- D7

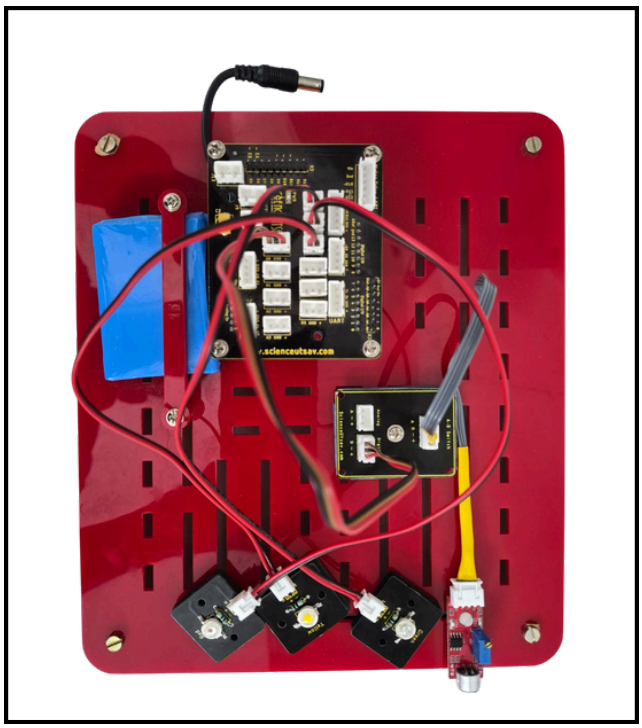
Step 1- Connect the AD switch to the board and attach the sound sensor to it



Step 2- Connect the servomotor as shown



Step 3- Connect the three LEDs



Step 4- Attach the template to the stand and fix the LEDs as shown

Code

```
// Session 2 Grade 9
// Sound Level Meter
// For RGB LED project, D12 is always HIGH

const int soundSensor = A0;

const int redLED = 10;
const int yellowLED = 11;
const int alwaysHighPin = 12;

void setup() {
 pinMode(redLED, OUTPUT);
 pinMode(yellowLED, OUTPUT);
 pinMode(alwaysHighPin, OUTPUT);

 digitalWrite(alwaysHighPin, HIGH); // D12 always HIGH
}

void loop() {
 int soundValue = analogRead(soundSensor);

 digitalWrite(alwaysHighPin, HIGH); // keep D12 always HIGH

 if (soundValue < 450) {
 digitalWrite(redLED, LOW);
 digitalWrite(yellowLED, LOW);
 } else if (soundValue < 700) {
 digitalWrite(redLED, LOW);
 digitalWrite(yellowLED, HIGH);
 } else {
 digitalWrite(redLED, HIGH);
 digitalWrite(yellowLED, HIGH);
 }

 delay(100);
}
```